

# Practical 9

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

```
GNU nano 7.2
#include <stdio.h>

int search(int key, int frame_items[], int frame_occupied) {
    for (int i = 0; i < frame_occupied; i++)
        if (frame_items[i] == key)
            return 1;
    return 0;
}

int predict(int pages[], int frame_items[], int n, int index, int frame_occupied) {
    int res = -1, farthest = index;

    for (int i = 0; i < frame_occupied; i++) {
        int j;
        for (j = index; j < n; j++) {
            if (frame_items[i] == pages[j]) {
                if (j > farthest) {
                    farthest = j;
                    res = i;
                }
            }
            break;
        }
        if (j == n)
            return i;
    }

    if (res == -1)
        return 0;
    else
        return res;
}

int main() {
    int frames[10], pages[50];
    int n, f, i;
    int hits = 0, faults = 0;

    printf("Enter number of frames: ");
    scanf("%d", &f);

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for (i = 0; i < n; i++)
        scanf("%d", &pages[i]);
}
```

```

        return res;
    }

    int main() {
        int frames[10], pages[50];
        int n, f, i;
        int hits = 0, faults = 0;

        printf("Enter number of frames: ");
        scanf("%d", &f);

        printf("Enter number of pages: ");
        scanf("%d", &n);

        printf("Enter page reference string:\n");
        for (i = 0; i < n; i++)
            scanf("%d", &pages[i]);

        int frame_occupied = 0;

        for (i = 0; i < n; i++) {
            if (search(pages[i], frames, frame_occupied)) {
                hits++;
                continue;
            }

            if (frame_occupied < f) {
                frames[frame_occupied++] = pages[i];
            } else {
                int pos = predict(pages, frames, n, i + 1, frame_occupied);
                frames[pos] = pages[i];
            }

            faults++;
        }

        printf("\nTotal Page Hits: %d", hits);
        printf("\nTotal Page Faults: %d", faults);
        printf("\nHit Percentage: %.2f%%", (hits * 100.0) / n);
        printf("Fault Percentage: %.2f%%\n", (faults * 100.0) / n);

        return 0;
    }
}

```

```

sh@DESKTOP-0TK9CH3:~$ nano optimal.c
sh@DESKTOP-0TK9CH3:~$ gcc optimal.c -o optimal
sh@DESKTOP-0TK9CH3:~$ ./optimal
Enter number of frames: 3
Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5

Total Page Hits: 5
Total Page Faults: 7
Hit Percentage: 41.67% Fault Percentage: 58.33%
sh@DESKTOP-0TK9CH3:~$ |

```